

Matgeo Presentation - Problem 4.7.51

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Problem Statement

Find the equation of the plane passing through the point $(-1, 3, 2)$ and perpendicular to each of the planes $x + 2y + 3z = 5$ and $3x + 3y + z = 0$.

Solution: Normal Vectors

The normal vectors to the given planes are $\mathbf{n}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\mathbf{n}_2 = \begin{pmatrix} 3 \\ 3 \\ 1 \end{pmatrix}$.

Let the normal vector of the required plane be $\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix}$. Since the required plane is perpendicular to both given planes, its normal vector $\mathbf{n}$ must be orthogonal to both $\mathbf{n}_1$ and $\mathbf{n}_2$.

Solution: System of Equations

This gives the following homogeneous system of linear equations:

$$\mathbf{n}_1^\top \mathbf{n} = n_1 + 2n_2 + 3n_3 = 0 \quad (0.1)$$

$$\mathbf{n}_2^\top \mathbf{n} = 3n_1 + 3n_2 + n_3 = 0 \quad (0.2)$$

To solve this system, we form the augmented matrix and reduce it. The final reduced row echelon form is:

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 3 & 3 & 1 & 0 \end{array} \right) \xrightarrow{\text{Row Reduce}} \left(\begin{array}{ccc|c} 1 & 0 & -7/3 & 0 \\ 0 & 1 & 8/3 & 0 \end{array} \right) \quad (0.3)$$

Solution: Finding the Normal Vector

From the reduced form, we have:

$$n_1 - \frac{7}{3}n_3 = 0 \implies n_1 = \frac{7}{3}n_3$$

$$n_2 + \frac{8}{3}n_3 = 0 \implies n_2 = -\frac{8}{3}n_3$$

The solution space for $\mathbf{n}$ is spanned by the vector $\begin{pmatrix} 7/3 \\ -8/3 \\ 1 \end{pmatrix}$. We can choose any non-zero scalar multiple of this vector for our normal. Letting the free variable $n_3 = -3$ gives a simple integer vector:

$$\mathbf{n} = \begin{pmatrix} -7 \\ 8 \\ -3 \end{pmatrix} \tag{0.4}$$

Solution: Final Equation

The equation of the required plane passes through the point $\mathbf{P}_0 = \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}$ and has the normal vector $\mathbf{n}$. The equation is $\mathbf{n}^\top \mathbf{p} = \mathbf{n}^\top \mathbf{P}_0$.

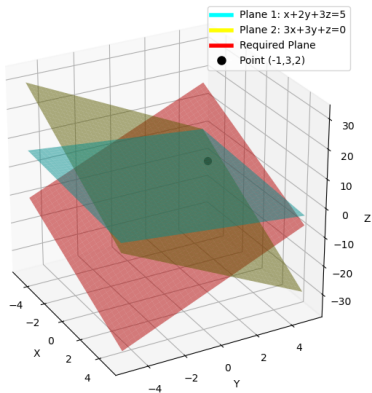
$$c = (-7 \quad 8 \quad -3)^\top \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix} = 7 + 24 - 6 = 25 \quad (0.5)$$

Thus, the equation of the plane is:

$$(7 \quad -8 \quad 3) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = -25 \quad (0.6)$$

Plot

Intersection of Planes and Required Perpendicular Plane



Code: solution.h

```
#ifndef PLANE_H
#define PLANE_H

#include <stdio.h>

// Function to compute cross product of two vectors
void cross_product(double a1, double b1, double c1,
                  double a2, double b2, double c2,
                  double *rx, double *ry, double *rz) {
    *rx = b1*c2 - c1*b2;
    *ry = c1*a2 - a1*c2;
    *rz = a1*b2 - b1*a2;
}

// Function to compute dot product of two vectors
double dot_product(double a1, double b1, double c1,
                  double a2, double b2, double c2) {
    return a1*a2 + b1*b2 + c1*c2;
}
```

C Code: solution.c

```
#include "solution.h"

int main() {
    double x0, y0, z0;
    printf("Enter point (x0 y0 z0): ");
    scanf("%lf%lf%lf", &x0, &y0, &z0);

    double a1, b1, c1, d1;
    printf("Enter coeffs of plane1 (a1 b1 c1 d1): ");
    scanf("%lf%lf%lf%lf", &a1, &b1, &c1, &d1);

    double a2, b2, c2, d2;
    printf("Enter coeffs of plane2 (a2 b2 c2 d2): ");
    scanf("%lf%lf%lf%lf", &a2, &b2, &c2, &d2);

    double n1x=a1, n1y=b1, n1z=c1;
    double n2x=a2, n2y=b2, n2z=c2;

    double nx, ny, nz;
    cross_product(n1x, n1y, n1z, n2x, n2y, n2z, &nx, &ny, &nz);
```

Python Code: solution.py

```
import numpy as np

def plane_equation(normal, point):
    d = np.dot(normal, point)
    return (*normal, d)

def main():
    x0, y0, z0 = map(float, input("Enter_point:_").split())
    a1, b1, c1, d1 = map(float, input("Enter_plane1_coeffs:_").split())
    a2, b2, c2, d2 = map(float, input("Enter_plane2_coeffs:_").split())

    n1 = np.array([a1, b1, c1])
    n2 = np.array([a2, b2, c2])

    n_required = np.cross(n1, n2)
    p = np.array([x0, y0, z0])

    a, b, c, d = plane_equation(n_required, p)
```

Python Code: plot8.py

```
import numpy as np
import matplotlib.pyplot as plt

def plane1_z(x, y): return (5-x-2*y)/3
def plane2_z(x, y): return -3*x - 3*y
def required_z(x, y): return (-7*x+8*y-25)/3

x = np.linspace(-5, 5, 40)
y = np.linspace(-5, 5, 40)
X, Y = np.meshgrid(x, y)
Z1, Z2, Zr = plane1_z(X,Y), plane2_z(X,Y), required_z(X,Y)
fig = plt.figure(figsize=(10, 7))
ax = fig.add_subplot(111, projection='3d')
ax.plot_surface(X, Y, Z1, alpha=0.5, color='cyan', label='x+2y+3z=5')
ax.plot_surface(X, Y, Z2, alpha=0.5, color='yellow', label='3x+3y+z=0')
ax.plot_surface(X, Y, Zr, alpha=0.5, color='red', label='Required Plane')
ax.scatter(-1, 3, 2, color='black', s=50, label='Point(-1, 3, 2)')
```